

Curriculum Vitae - Javier Hernández Rodríguez

Personal Information

Name	Javier Hernández Rodríguez
Affiliation	 Departamento de Química Física, Plaza de los Caídos s/n, Universidad de Salamanca, Salamanca, Spain.
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Education

03/2022-present

PhD in Chemical Science and Technology, Universidad de Salamanca.

Supervisors: Dr. Susana Gómez Carrasco (U. de Salamanca)
Dr. Cristina Sanz-Sanz (U. Autónoma de Madrid)

09/2023-06/2024

Master's Degree in Secondary Education Teaching. Universidad de Salamanca.

09/2020-09/2022

Erasmus Mundus Master in Theoretical Chemistry and Computational Modelling. Universidad Autónoma de Madrid.

Thesis title: Quantum dynamics of spin-forbidden transitions.

Supervisors: Dr. Cristina Sanz-Sanz (U. Autónoma de Madrid)
Dr. Susana Gómez Carrasco (U. de Salamanca)

09/2013-01/2020

Bachelor in Chemistry. Universidad Autónoma de Madrid.

Undergraduate project title: Cálculo y ajuste de las Superficies de Energía Potencial del sistema LiH_2^+ . Calculation and fitting of Potential Energy Surfaces of LiH_2^+ system.

Supervisor: Prof. Miguel Paniagua (U. Autónoma de Madrid)

Research Experience - Positions

09/2025-present

Fixed-term Teaching Fellowship, Universidad de Salamanca.

Department: Physical Chemistry

09/2024-09/2025

Fixed-term Teaching Fellowship, Universidad de Salamanca.

Department: Physical Chemistry

03/2022-04/2024

Researcher (Predoctoral contract), Universidad de Salamanca.

Department: Physical Chemistry.

Short Research Visiting Periods

03/2026-04/2026 Université Paris-Saclay, Orsay, France

Supervisor: David Lauvergnat

Research activity: Development of automatic kinetic energy operators for oxygen reactivity dynamics.

02/2026-03/2026 Université Bourgogne, Dijon, France.

Supervisor: Fabien Gatti

Research activity: Implementation of polyspherical coordinates for multi-dimensional MCTDH wavepacket propagations.

01/2025-02/2025 Consiglio Nazionale delle Ricerche, Pisa, Italy

Supervisor: Fabrizio Santoro

Research activity: Construction of Linear Vibronic Coupling Hamiltonian for tetraphenylpyrazine dimer.

Research Skills and Expertise

Electronic Structure

Software: MOLPRO, GAUSSIAN, ORCA, MOLCAS, Q-CHEM.

Methods: CASSCF, MRCI, (PNO-)CASPT2, CCSD(T), (TD-)DFT.

Molecular Dynamics, Potential Energy Surface fitting

QUANTICS: MCTDH, ML-MCTDH wavepacket propagation.

SHARC: Surface-Hopping dynamics, LVC Hamiltonians.

OVERDIA: Fragment diabaticization.

Programming and other software

Bash, Fortran, Gnuplot, Python, LaTeX, Mathematica, Chemcraft, VMD, GaussView.

Peer-Reviewed Publications

2026

J. Hernández-Rodríguez, A. Martín Santa Daría, S. Gómez-Carrasco, S. Gómez. Ultrafast nonadiabatic dynamics of tetraphenylsubstituted nitrogen-based heterocycles. *Small Struct.*, **7**, e70567 (2026) [Link](#)

J. Hernández-Rodríguez, S. Gómez-Carrasco, C. Sanz-Sanz and P.G. Jambrina. Quantum scattering pathway of the cofactorless spin-forbidden O₂ addition to DPA-CoA. *J. Chem. Phys.*, **165**, 024311 (2026) [Link](#)

2025

Léon L. E. Cigrang, Basile F. E. Curchod, Rebecca A. Ingle, Aaron Kelly, Jonathan R. Mannouch, Davide Accomasso, Alexander Alijah, Mario Barbatti, Wiem Chebbi, Nadja Došlić, Elliot C. Eklund, Sebastian Fernandez-Alberti, Antonia Freibert, Leticia González, Giovanni Granucci, Federico J. Hernández, J. Hernández-Rodríguez, Amber Jain, Jiří Janoš, Ivan Kassal, Adam Kirrander, Zhenggang Lan, Henrik R. Larsson, David Lauvergnat, Brieuc Le Dé, Yeha Lee, Neepa T. Maitra, Seung Kyu Min, Daniel Peláez, David Picconi, Zixing Qiu, Umberto Raucci, Patrick Robertson, Eduarda Sangiogo Gil, Marin Sapunar, Peter Schürger, Patrick Sinnott, Sergei Tretiak, Arkin Tikku, Patricia Vindel-Zandbergen, Graham A. Worth, Federica Agostini*, Sandra Gómez*, Lea M. Ibele*, and Antonio Prlj*. Roadmap for Molecular Benchmarks in Nonadiabatic Dynamics. *J. Phys. Chem. A*, **129**(31), 7023-7050 (2025). [Link](#)

2024

J. Hernández-Rodríguez, A. Martín Santa Daría, M. Soledad Alquegui, L. González-Sánchez, S. Gómez*. Role of dark states and Stokes shift simulations for tetraphenylpyrazine compared to other donor-acceptor photosensitizers. *ChemPhysChem*, e202400563 (2024). [Link](#)

A. Martín Santa Daría, J. Hernández-Rodríguez, L. M. Ibele, S. Gómez*. Photofragmentation of cyclobutanone at 200 nm: TDDFT vs CASSCF electron diffraction. *J. Chem. Phys.*, **160**, 114303 (2024). [Link](#)

2023

J. Hernández-Rodríguez, C. Sanz-Sanz, P. A. Enríquez, M. González*, M. Paniagua*. Potential energy surfaces for singlet and triplet states of the LiH₂⁺ system and quasi-classical trajectory cross sections for H + LiH⁺ and H⁺ + LiH.

Phys. Chem. Chem. Phys., **25**(41), 28052-28062 (2023). [Link](#)

Dissemination Articles

J. Hernández-Rodríguez, A. Veselinova, A. M. Santa-Darí, L. González-Sánchez and P. G. Jambrina*. Práctica de laboratorio computacional: Uso de cálculos de estructura electrónica para explicar la libre rotación de enlaces CC y las estructuras de Newman. *GALICIA QUÍMICA*, **1**(7), 29-38 (2025). [Link](#)

Research Projects

2025

Proyectos de Generación de Conocimiento 2024

Reference: PID2024-155352NB-C22. PIs: Manuel Lara Garrido, Alexandre Zanchet.

Title: Estructura Electrónica, colisiones moleculares y procesos ultrarápidos fotoinducidos.

2024

Proyectos de Generación de Conocimiento 2023

Reference: PID2023-147215NB-I00. PI: Pablo García Jambrina.

Title: Dinámica de Colisiones Reactivas e Inelásticas en fase gas y de procesos de incorporación de oxígeno molecular en sistemas biológicos.

Programa C1, University of Salamanca. PI: S. Gómez

Title: Mecanismos de cruce entre sistemas en materiales orgánicos para fotoemisión.

2021

Proyectos de Generación de Conocimiento 2020.

Reference: PID2020-113147GA-I00. PI: Pablo García Jambrina

Title: Dinámica Adiabática y no Adiabática de Reacciones y Cruces Intersistema en Sistemas Biológicos (INTERDYN)

Attended Conferences and Workshops

2026

Deciphering Spin–Orbit-driven O₂ addition to DPA–CoA through quantum scattering **(Poster)**. *HDQD2026 – High Dimensional Quantum Dynamics Conference, Salamanca, Spain*.

2025

Unraveling the effect of spin-orbit coupling in DpgC-catalyzed peroxidation **(oral)**. *VII Jornadas de Jóvenes Investigadores en Ciencia y Tecnología Químicas, Universidad de Salamanca, Salamanca, Spain*.

Non-Adiabatic Dynamics of Excited States in Tetraphenylpyrazine: From Monomer to Dimer **(invited talk)**. *Third General Meeting COSY Cost Action, Bologna, Italy*.

Photoinduced Nonadiabatic Processes in Tetraphenyl-Substituted Azaheterocycles **(oral)**. *15th Symposium on Computing π -conjugated compounds, Università di Siena, Siena, Italy*.

2024

Quantum dynamics of the spin-forbidden peroxidation catalyzed by the DpgC enzyme **(oral)**. *13th International Meeting on Atomic and Molecular Physics and Chemistry (IMAMPC), University of Warsaw, Warsaw, Poland*.

N-doped PAH non-adiabatic dynamics with the ML-MCTDH method **(invited talk)**. *Second General Meeting COSY Cost Action, Wroclaw, Poland*.

Insights into quantum dynamics of DpgC-catalyzed peroxidation **(poster)**. *Segundo Simposio de Química Física y Transferencia de Conocimiento, Universidad Autónoma de Madrid, Madrid, Spain*.

Quantum effects of the peroxidation of organic substrates in absence of cofactors **(oral)**. *V Jornadas de Jóvenes Investigadores en Ciencia y Tecnología Químicas, Universidad de Salamanca, Salamanca, Spain*.

Cyclopropanone as a benchmark system for non-adiabatic dynamics **(flash talk)**. *CECAM workshop. Standardizing*

nonadiabatic dynamics: towards common benchmarks, Paris, France.

A quantum dynamics study of tetraphenylpyrazine molecular crystals **(poster)**. *CECAM workshop. Theoretical and Experimental Advances in Atmospheric Photochemistry, CECAM, Lausanne, Switzerland.*

A step towards quantum dynamics in tetraphenylpyrazine molecular crystals **(oral)**. *15th Young Researchers in Atomic and Molecular Physics Conference, RSEQ, Valencia, Spain.*

2023

Intersystem crossing dynamics of O₂ and a DPA-CoA model **(oral)**. *3rd Conferencia Jóvenes Investigadores, FJI-INNOVA, Salamanca, Spain.*

Unravelling the role of quantum effects in the peroxidation of organic substrates **(poster)**. *Multiscale modeling of the properties of compounds, COSY Cost Action, Belgrade, Serbia.*

The singlet-triplet reaction in a enzymatic peroxidation: a quantum dynamics study **(poster)**. *Training School: Theory and modeling of dynamics of molecules and clusters in the gas phase, MD-GAS Cost Action, Gdansk, Poland.*

Potential energy surfaces and quasi-classical trajectory study of the LiH₂⁺ system **(poster)**. *Primer Simposio de Química Física, Universidad Autónoma de Madrid, Madrid, Spain.*

Quantum effects of the enzymatic peroxidation of organic substrates: a quantum dynamic study **(poster)**. *V Jornadas de Jóvenes Investigadores en Ciencia y Tecnología Químicas, Universidad de Salamanca, Salamanca, Spain.*

Quantum effects of the enzymatic peroxidation of organic substrates without cofactors **(flash talk)**. *XXXIX Reunión Bienal de la RSEQ, Zaragoza, Spain.*

Quantum effects of the enzymatic peroxidation of organic substrates **(poster)**. *CECAM Workshop. Triggering out-of-equilibrium dynamics in molecular systems, CECAM, Switzerland.*

Quantum effects of forbidden spin transitions in reactions between organic substrates and molecular oxygen **(oral)**. *14th Young Researchers in Atomic and Molecular Physics Conference, RSEQ, Madrid, Spain.*

2022

Quantum dynamics of the addition of molecular oxygen to organic substrates in absence of cofactors **(poster (Award))**. *Dynamics of Energetic and Electronic Processes in molecules and clusters in the GAS phase, MD-GAS Cost Action CA18212, Madrid, Spain.*

Quantum effects of the peroxidation of organic substrates in absence of cofactors **(oral)**. *IV Jornadas de Jóvenes Investigadores en Ciencia y Tecnología Químicas, Universidad de Salamanca, Salamanca, Spain.*

Quantum dynamics of the addition of O₂ to organic substrates **(poster)**. *High Dimensional Quantum Dynamics (HDQD), University of Groningen, Groningen, Netherlands.*

2019

Potential energy surfaces for singlet and triplet states of LiH₂⁺ system **(poster)**. *International Meeting on Atomic and Molecular Physics and Chemistry (IMAMPC), UAM-CSIC, Madrid, Spain.*

Conference Organization

2026

Electronic Structure Principles and Applications(ESPA2026), Universidad de Salamanca, Salamanca, Spain.
(Organizing Committee) [Link](#)

Teaching

Physical Chemistry I (Quantum Chemistry). Degree Program: Chemistry (2nd Year)

2025-2026: Computational laboratory sessions (27 hours).

2024-2025: Computational laboratory sessions (105 hours) and problem-solving sessions (30 hours).

Physical Chemistry Laboratory. Degree Program: Chemistry (3rd Year)

2025-2026: Laboratory sessions (64 hours).

Instrumental Techniques. Degree Program: Pharmacy (1st Year)

2025-2026: Laboratory sessions (45 hours).

2024-2025: Laboratory sessions (75 hours).

Thermodynamics and Chemical Kinetics. Degree Program: Pharmacy and Biotechnology (1st Year)

2025-2026: Laboratory sessions (60 hours).

2024-2025: Laboratory sessions (30 hours).

Physical Chemistry. Degree Program: Chemical Engineering (1st Year)

2025-2026: Laboratory sessions (16 hours).

Applied Thermodynamics. Degree Program: Chemical Engineering (2nd Year)

2025-2026: Laboratory sessions (16 hours).

Chemical Kinetics. Degree Program: Chemical Engineering (3rd Year)

2025-2026: Laboratory sessions (12 hours).